

Exploratory Analysis of TANGEDCO Electricity Demand

Electricity Utility & Power Sector

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DAILY LOAD • WEATHER •
SEASONAL PATTERNS

Study window
01 Jan 2021 – 27 Apr 2024

Industry Overview

POWER & ELECTRICITY UTILITIES

Electricity systems must balance changing demand with reliable supply. Tamil Nadu demand is shaped by household use, industry, commercial activity, weather and seasonal behaviour.

- Peak-load monitoring helps utilities understand periods of maximum system stress.
- Daily energy-supplied data provides a view of electricity delivered.
- Weather and calendar patterns can support planning, forecasting and resource allocation.

WHY ANALYSE THE DATA?

- 01 Reliability**
Understand high-load periods
- 02 Planning**
Compare years and months
- 03 Evidence**
Explore load-weather relationships

Problem Statement

Electricity demand changes over time. Without structured analysis, it is harder to identify recurring high-load periods and understand how demand aligns with weather and energy supplied.

THE CHALLENGE

Find demand patterns

Daily readings are noisy; year-wise and month-wise comparisons reveal the broader pattern.

WHY IT MATTERS

Plan for peak periods

Peak demand can influence capacity planning, operational readiness and demand-side measures.

WHO IS AFFECTED

Utilities and consumers

Grid operators, planners, businesses and households depend on reliable electricity.

Project focus: describe trends and relationships—not claim causation or forecast future demand.

Project Objectives

01

Prepare a consistent dataset

Clean and align daily electricity and weather records for the common study period.

02

Describe demand over time

Examine daily trends, yearly averages, monthly seasonality and day-of-week patterns.

03

Explore weather relationships

Compare peak load with temperature, solar radiation, UTCI and cloud cover.

04

Identify high-load days

Rank days by peak load and review associated energy and temperature values.

05

Communicate evidence clearly

Use summary statistics and charts to support practical, data-led recommendations.

Proposed Solution & Analysis Questions

PROPOSED SOLUTION

A Python EDA workflow

- Load electricity and weather files
- Clean dates, duplicates and numeric columns
- Filter to the shared study period
- Merge on Date and engineer calendar/weather features
- Summarise, visualise and interpret

TOOLS

Pandas • NumPy • Matplotlib • Seaborn

QUESTIONS THIS PROJECT ANSWERS

- How did daily peak load change from 2021 through April 2024?
- Which years and months had the highest average peak load?
- Which days recorded the largest peak-load values?
- How strongly does peak load relate to daily energy supplied?
- How are temperature, solar radiation, UTCI and cloud cover associated with peak load?
- What does the recorded unmet-demand field show?

Dataset Information

1,195

Merged daily rows

Common dates after inner joins

25

Merged columns

Before feature-engineering additions

2021–2024

Study period

Through 27 April 2024

CSV

File format

Daily tabular data

DATASET / SOURCE

Daily electricity demand data for Indian states (2014–2024)

Zenodo record: 10.5281/zenodo.14983362
Electricity series scraped from India's national grid; daily Tamil Nadu weather variables are joined by date.

IMPORTANT FEATURES

- Date
- Peak_Load_MW
- Energy_Supplied_MU
- Unmet_Demand_MW
- Temperature, solar radiation, cloud cover, UTCI

Data Collection & Understanding

Four source tables were loaded in the notebook, inspected with shape/head/column checks, then filtered and cleaned before joining Tamil Nadu electricity values to weather data.

Source table	Raw rows × cols	Cleaned rows × cols	Missing after cleaning
Peak load	3,955 × 40	1,196 × 40	3,782
Peak unmet demand	3,955 × 40	1,196 × 40	3,782
Daily energy supplied	3,955 × 40	1,196 × 40	3,782
Tamil Nadu weather	16,554 × 22	1,213 × 22	291

INITIAL DATA CHECKS

- Date column parsed as datetime.
- Non-date fields converted to numeric where possible.
- Duplicate dates removed; cleaned tables report 0 duplicate dates.
- Weather data contains missing values, including some UTCI observations.
- Inner join produced 1,195 records and 25 columns.

Missing counts for electricity source tables include other states' columns; the final merged table focuses on Tamil Nadu plus weather.

Data Cleaning & Transformation

1

Parse dates

Rename date field to Date and convert to datetime.

2

Filter dates

Target 2021-01-01 to 2024-04-28.

3

Deduplicate

Drop invalid dates and duplicate dates; sort chronologically.

4

Convert types

Convert non-date fields to numeric; invalid values become missing.

5

Select & merge

Select Tamil Nadu values and inner-join all tables on Date.

6

Engineer features

Add year/month/quarter/day/weekday; convert temperature K→°C; derive unmet-demand ratio and energy-per-peak indicator.

Final merged table: 1,195 rows × 25 columns • actual merged range: 2021-01-01 to 2024-04-27

Exploratory Data Analysis (EDA)

15,221.78 MW

Mean daily peak load

Across 1,195 merged dates

20,583 MW

Maximum peak load

26 April 2024

323.88 MU

Mean daily energy supplied

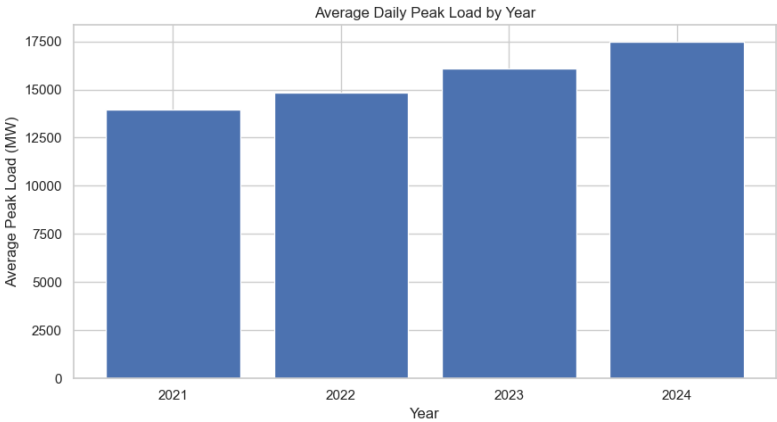
Daily series summary

0 MW

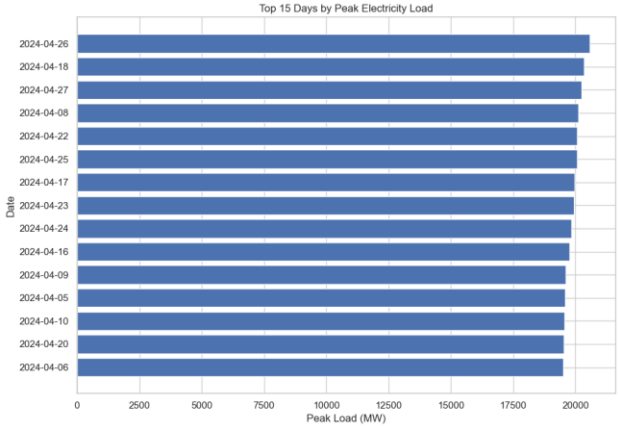
Recorded unmet demand

All selected records are zero

AVERAGE PEAK LOAD BY YEAR



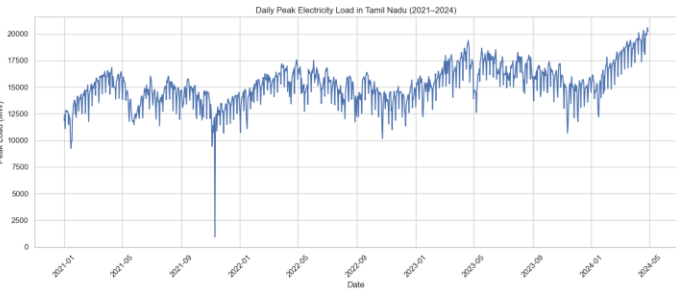
TOP 15 PEAK-LOAD DAYS



EDA methods: Pandas descriptive statistics and groupby summaries; NumPy-derived indicators; IQR-based outlier review.

Data Visualization

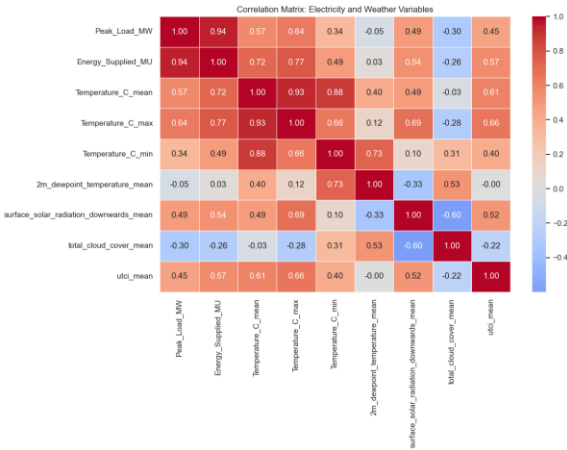
DAILY PEAK-LOAD TREND



Purpose: see change over time.

Insight: peak load rises over the study window; the highest observed value occurs in April 2024.

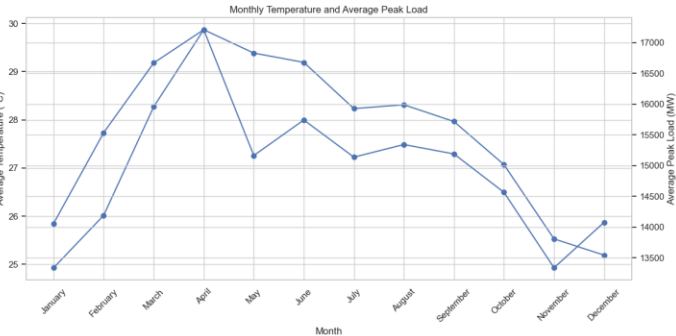
CORRELATION HEATMAP



Purpose: compare linear associations.

Insight: peak load correlates with energy supplied ($r \approx 0.94$); max temperature is moderately positive ($r \approx 0.64$).

MONTHLY TEMP & PEAK LOAD



Purpose: compare seasonal patterns.

Insight: average peak load is highest in April and lowest in November. Association is not proof of causation.

Other notebook charts include yearly energy supplied, month-by-year heatmap, weekday comparison, temperature scatter plots and load distribution.

Key Insights & Recommendations

KEY INSIGHTS FROM THE DATA

- Annual mean peak load rose from 13,955.53 MW (2021) to 16,087.79 MW (2023). 2024 is partial-year.
- Highest observed daily peak: 20,583 MW on 26 Apr 2024; 442.3 MU supplied that day.
- April had the highest average monthly peak load (17,211.97 MW); November the lowest (13,337.99 MW).
- Peak load correlation: energy supplied $r \approx 0.944$; maximum temperature $r \approx 0.643$.
- Unmet_Demand_MW is zero for all 1,195 merged records; this field cannot explain sample variation.

RECOMMENDATIONS

- Prioritise closer monitoring during April and other high-load periods.
- Combine longer demand history with weather and calendar variables for future forecasting.
- Validate the all-zero unmet-demand field against an independent official source.
- Document missing-value treatment and preserve raw files for reproducibility.
- Investigate peak days with additional operational context before proposing interventions.

Correlation indicates association, not causation. Electricity data are national-grid scraped series; this is a Tamil Nadu-focused analysis, not a direct TANGEDCO publication.

CONCLUSION

From daily records to planning-relevant patterns

The project assembled a date-aligned dataset of electricity and weather variables, applied cleaning and feature engineering, and used EDA to describe load trends, seasonal variation and weather associations.

FUTURE SCOPE

Add verified official records, investigate missing values, extend the time series, and test forecasting models only after data-quality checks.

WHAT WAS ACHIEVED

- 1,195 merged daily records
- 25 original merged columns before engineered features
- Peak load maximum: 20,583 MW
- Year/month and weather relationships explored
- Data-quality caveat identified for unmet demand